

6. FUNCTIONAL AND PERFORMANCE TESTING

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Date	01 October 2026
Team ID	SWTID-2026-4302
Project Name	Import Data using Transform Maps (Spreadsheet)
Team Members	Prasanna K, Yousuf Riyas S, Ratnakar K S, Mohammed Thoufiq I

6.1 Performance Testing

Testing covers three areas: **functional testing** of each import step, **performance testing** of the import run, and **User Acceptance Testing (UAT)** of the finished solution. Results marked Pass come from the import runs carried out in the project; cases not yet executed are marked **Not tested** and can be completed by the team.

6.1.1 Test Scenarios and Results

Test Case ID	Scenario	Test steps	Expected result	Actual result	Pass / Fail	Tester
TC-001	Upload spreadsheet with Load Data	System Import Sets > Load Data; select Employee Import; choose the .xlsx file; Sheet 1, Header row 1; Submit	Rows load into u_employee_import; import shows Completed / Success	Import completed successfully	Pass	Yousuf Riyas S
TC-002	Field mapping (Auto Map Matching Fields)	Open Sample Spreadsheet Import; click Auto Map Matching Fields; Save; review Field Maps	Employee ID, Name, Email, Department, Location map to Employee Test fields	Transform ran and populated the target table	Pass	Ratnakar K S
TC-003	Run transform	Click Transform (Run Transform); select the map; Transform	Success message; state Complete	Transform completed with success message	Pass	Ratnakar K S
TC-004	Verify target table	Open Employee Test table from the Application Navigator	All imported records are listed with correct field values	All records created in Employee Test	Pass	Yousuf Riyas S

Test Case ID	Scenario	Test steps	Expected result	Actual result	Pass / Fail	Tester
TC-005	Coalesce updates and inserts	Set Coalesce = true on Employee ID; load a 4-row sheet (2 existing IDs with changed Name/Email, 2 new IDs); run transform; open Transform History	2 rows updated, 2 rows inserted	2 inserted, 2 updated (e.g. email test10 → test18; name Ajay kumar → Ajay)	Pass	Mohammed Thoufiq I
TC-006	Re-import identical sheet (no duplicates)	Load the same 4-row sheet again; run transform; open Transform History	0 inserted, 0 updated, 4 ignored; no duplicate records	0 inserted, 0 updated, 4 ignored	Pass	Mohammed Thoufiq I
TC-007	Reports and dashboard	Create 3 reports on Employee Test; create Employee Analytics Dashboards; add the reports	Dashboard shows the three reports	Three reports added to the dashboard	Pass	Prasanna K
TC-008	Validate required fields	Load a row with a blank Employee ID, Name or Email; run transform	Incorrect or incomplete row is identified for correction	Not yet tested	Not tested	Mohammed Thoufiq I

6.1.2 Transform History Results (Coalesce Test)

Run	Data loaded	Total rows	Inserted	Updated	Ignored
Run 1	Updated sheet: 2 existing IDs with changed Name/Email + 2 new IDs	4	2	2	0
Run 2	Same sheet loaded again	4	0	0	4

Changes tested: one employee's (Raviteja) email changed from test10@gmail.com to test18@gmail.com; Employee ID SB-0004 name changed from "Ajay kumar" to "Ajay"; two new Employee IDs added. In Transform History, an "ignored" row means no field values changed, so no record was updated or created.

6.1.3 Performance Test Cases

Test Case ID	Scenario	Test steps	Expected result	Status
PT-01	Import run time (sample sheet)	Run the transform and read Run time in Transform History	Completes without errors or timeout	Not tested – record run time

Test Case ID	Scenario	Test steps	Expected result	Status
PT-02	Larger file load	Load a sheet with several hundred rows and run the transform	All rows processed; counts in Transform History add up to the row total	Not tested

Metrics to record for performance testing: rows loaded, transform run time, and inserted / updated / ignored / skipped / error counts per run.

6.1.4 Report and Dashboard Testing

S.No.	Parameter	Details
1	Data Rendered	Employee Test table (u_employee_test) lists all imported records
2	Data Preprocessing	Import Set table → Transform Map (Auto Map Matching Fields) → target table
3	Utilisation of Filters / Grouping	Group by Department (pie chart), group by Location
4	Calculation / Aggregation used	Aggregation: Count
5	Dashboard design	No. of visualisations / graphs: 3 reports on Employee Analytics Dashboards (Department, Location, Employee List)

6.2 User Acceptance Testing (UAT)

Purpose of document

This section explains the test coverage and open issues of the project at the time of release to User Acceptance Testing.

Project version	1.0
Testing scope	Spreadsheet preparation and table creation; Load Data, Transform Map and transform execution; Coalesce on Employee ID; Reports and Employee Analytics Dashboards; user stories USN-1 to USN-15
Testing environment	ServiceNow Personal Developer Instance (credentials shared separately; do not record passwords here)
Testing period	

Test case analysis

Section	Total cases	Not tested	Fail	Pass
Data Loading (TC-001)	1	0	0	1
Transform Map (TC-002, TC-003)	2	0	0	2
Target Data Verification (TC-004)	1	0	0	1
Coalesce / Duplicate Prevention (TC-005, TC-006)	2	0	0	2
Reports & Dashboard (TC-007)	1	0	0	1
Validation of Required Fields (TC-008)	1	1	0	0
Totals	8	1	0	7

Defect analysis

No defects had been logged at the time of writing. Complete the table if defects are found during UAT (TC-008 validation is the main open item).

Resolution	Severity 1	Severity 2	Severity 3	Severity 4	Subtotal
By Design	-	-	-	-	-
Duplicate	-	-	-	-	-
External	-	-	-	-	-
Fixed	-	-	-	-	-
Not Reproduced	-	-	-	-	-
Skipped	-	-	-	-	-
Won't Fix	-	-	-	-	-
Totals	-	-	-	-	-

Bug tracking

Bug ID	Description	Steps to reproduce	Severity	Status	Additional feedback
BG-001	No defects recorded so far	-	-	-	Add issues found during UAT (for example TC-008 validation)

Sign-off

Role	Name	Date	Signature
Tester	Mohammed Thoufiq I		
Project manager / Team lead	Prasanna K		
Product owner			